AS MATH 9709 · EXTRA CREDIT · TRIMESTER 1

Teach It in a Video

Teach it. Film it. Boost your grade.

+5
points on your T1 grade

Mon Nov 2	Mon Nov 9	Mon Nov 16	Thu Nov 19
Topic sign-up opens	Own-idea deadline	Video due, 11:59 pm	Last day to fix and resubmit

You have been learning this all trimester. Now you get to be the teacher. Make a short video that teaches one idea from AS Math, so good that a classmate could learn it from you.

A top video raises your T1 grade by up to 5 points. Want to pull your grade up before the trimester ends? This is your chance.

The must-haves

Required

1. **One idea.** From the topic menu, or your own idea approved by Mon Nov 9.
2. **Correct math.** Cambridge notation, exact answers or 3 significant figures, and say degrees or radians.
3. **A worked example.** At least one, start to finish, with the working shown. No jumping to a calculator answer.
4. **30 seconds to 3 minutes.** Any platform shape: vertical or wide.
5. **Privacy rules.** Every rule in the Stay Safe box.
6. **A complete comment.** Video type (TikTok, Reel, Short, YouTube video, or other), topic, group and parts, credits, and the AI line.

Make it legendary

Ideas, not rules

- Hook us in the first 3 seconds.
- Show the mistake everyone makes, then fix it.
- End with a challenge question for the viewer.
- Connect it to real life: sports, music, money, games, St. Croix.
- Tell a story, act a skit, write a song, ride a trend, race the clock.
- Draw it, animate it, build it, act it out.
- Use a real Cambridge question (name it, like 9709/11 June 2023 Q3).
- If it has never been done, do it.

Teams

Go solo or team up with up to 2 classmates from this class (3 people max). Everyone in the group gets the same score. Each person uploads the video, and the comment says who did what.

Questions? Ask me in class or email akrigger@ghcds.org.

Any format works

Faceless voiceover
Screen recording or visuals with your voice.

Whiteboard or paper
Hands only, writing it out.

Illustrated or animated
Drawings, stop motion, animation apps.

On camera
You, your group, a skit, a challenge.

Demonstration
Real objects, Desmos visuals, a physical model.

AI-generated or designed
Slides, graphics, AI video. See the AI rule.

Quality and how well it teaches decide your score. The tool you use is up to you. No phone needed. A school laptop's camera or screen recorder works for every format. You can record in the lab during Day 1 academic time.

Posting it

- Post it if you want: TikTok, Instagram Reel, YouTube Short, or YouTube. Posting is optional and earns no extra points.
- Always upload the file to Schoology too. That is the copy that gets graded.
- Your video may be picked for a showcase. If it is, I will ask you and a parent first.

Stay safe Required

- First name only, or no name.
- No school uniform, logo, address, or anything that shows where you live.
- Nobody else appears or is heard unless they said yes.
- Music from your app's library, a royalty-free library, or none. Credit outside music and images.
- Keep it school appropriate.

Using AI

No deduction	AI explaining the idea while you learn it, checking an answer you already worked out, captions, and audio cleanup.
Minus 4 rubric points	Full AI assistance: AI wrote the script, did the voice or avatar, or generated the video.
Always	Write "AI used:" in your comment, and be ready to explain every step to me live.

Turn it in

1. Export your video as MP4 or MOV, under 512 MB. Too big? Export at 720p.
2. Open "Extra Credit: Teach It in a Video" in Schoology and click Submit Assignment.
3. Choose Upload (not Create) and pick your file.
4. In the comment write: the video type (TikTok, Instagram Reel, YouTube Short, YouTube video, or other), your topic, your group members and each person's part, music and image credits, and "AI used:" followed by what AI did, or "none." Posted it? Add the link.
5. Click Submit. In a group? Every member does steps 2 to 5.

Something wrong in your video? I send it back once. Fix it by Thursday, November 19 and it is graded like new.

Questions? Ask me in class or email akrigger@ghcds.org.

How it is scored 16 points

Criteria	4 · Nailed it	3 · Solid	2 · Getting there	1 · Not yet
Accurate	Everything is correct, with the right words and notation.	Correct, with one small slip that does not change the result.	A few slips, or words and notation that are off.	The main idea or answer is wrong. Sent back to fix.
Teaches clearly	A classmate could learn it from this. Every step makes sense.	Clear overall. One step is rushed.	Hard to follow in places.	Hard to follow most of the way.
Hook and creativity	Grabs attention right away. A fresh idea or style nobody else had.	Good start that holds attention.	Slow start, plain style.	No hook. Feels like reading a list.
Format and credits	30 seconds to 3 minutes, easy to see and hear, privacy rules kept, comment complete.	One small issue, like quiet audio.	Two issues, or credits or the AI line missing.	Way over or under time, or hard to see or hear.

Rubric score	T1 grade boost
16	**+5.0**
14	**+4.4**
12	**+3.8**
10	**+3.1**
9	**+2.8**
8 or less	**0**

What the boost means. Your rubric score turns into extra points on your overall T1 grade. A perfect 16 adds about 5 points, so an 84 becomes about an 89.

Each rubric point is worth about 0.3 on your T1 grade. Full AI assistance subtracts 4 rubric points first. Your T1 grade cannot go above 100, so if the boost would take you past 100, only the points up to 100 count. A 98 with a full boost ends at 100.

Breaking a Stay Safe rule means the video comes back and earns nothing until it is fixed.

Pick your topic

One person or group per topic, first come, first served, in the Schoology sign-up discussion.

1.1 Quadratics

- Completing the square and the vertex
- The discriminant: how many roots?
- Solving a quadratic inequality
- A line and a quadratic, solved together
- Disguised quadratics, like $x^4 - 5x^2 + 4 = 0$

1.2 Functions

- Domain and range
- Composite functions fg(x)
- Finding an inverse function
- Transformations of graphs

1.3 Coordinate geometry

- Parallel and perpendicular gradients
- Circle equation: center and radius
- Tangent to a circle
- Where a line meets a curve
- Chords and the angle in a semicircle

1.4 Circular measure

- Radians, and why we use them
- Arc length
- Sector area
- Segment area

Related ideas

- A quadratic in sports: a ball's path
- Radians in real machines: wheels and gears
- Your own idea (approved by Nov 9)

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